

Homework: Artificial Intelligence in Action - Applied Programming Project

Objective: The goal of this assignment is to bridge the gap between theoretical artificial intelligence concepts and practical software development. You are required to design, implement, and showcase a working software application utilizing at least one of the core algorithms covered in the course materials.

Project Topics (Choose One):

1. **Intelligent Game Agent:** Develop an AI that plays a two-player turn-based game (e.g., Tic-Tac-Toe) using the Minimax algorithm. You are highly encouraged to optimize your AI's computation time by implementing Alpha-Beta Pruning.
2. **Search & Pathfinding Visualizer:** Create an application that solves constraints like the "Rat in a Maze" problem or navigates a graph network. You must implement algorithms such as Breadth-First Search (BFS), Depth-First Search (DFS), or Dijkstra's Shortest Path Algorithm and demonstrate how they reach the goal.
3. **Optimization Simulation:** Build a tool that attempts to find the optimal solution to a complex problem avoiding local maximums. You can choose to implement Hill Climbing, Simulated Annealing, or a Genetic Algorithm (simulating natural selection through crossover and mutation).
4. **Probabilistic Modeling & Simulation:** Develop a Monte Carlo simulation to estimate complex values (such as estimating Pi) and demonstrate the Law of Large Numbers. Alternatively, implement a Markov matrix-based simulation (similar to the foundation of Google's PageRank) to calculate steady-state probabilities.

Technical Requirements:

- **Working Application:** Your application must be fully functional. While Python is highly recommended due to its rich libraries and course examples (e.g., scipy, numpy), you may use any programming language or technology stack you prefer. You can build a terminal-based app, a web app, or a desktop GUI.

Deliverables & Submission Guidelines: You must upload the following two items to the system by the deadline:

1. **Project Report (PDF Document):** Submit a comprehensive report explaining your chosen problem, the mathematical/theoretical background of the algorithm you used, and a brief overview of your code architecture.
2. **Demonstration Video (Max 2 Minutes):** Record a screencast demonstrating your working application. In this video, briefly explain how the algorithm runs in the background while showing the program in action. Upload the video to a platform (YouTube, Google Drive, etc.) and ensure the visibility is set to 'Unlisted' or 'Public'.
Submit the video link along with your report.